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Btech CSE Cou 'A'

Quick Work

Page No.:

Date:

M T W T F S S

Assignment - 04

DBMS Unit-04

Section-A

Ques-1 What is transaction in DBMS?

⇒ A transaction is a logical unit of work that consist of multiple databases operations (like INSERT, UPDATE, DELETE) which are executed together.

Example :- Bank Transfer (debit from one account + credit to another).

Ques-2 Explain ACID Properties.

⇒ ACID properties ensure reliable transaction processing.

1) Atomicity → A transaction is treated as a single-unit. In this Either all operations are executed or none are executed. If any step fails it entire transaction is rolled back.

2) Consistency → It ensures database moves from one valid state to another valid state.
↳ Here, all rules, constraines & relationships must be maintained.

3) Isolation → In this multiple transactions run independently without affecting each other

Quick Work
Page No.:
Date:
M T W T F S S

- In this Intermediate results are not visible to other transactions.

(4) Durability :- Once a transaction is committed changes are permanently stored.

- Data remains safe even after system failure.

Ques-3 What is concurrency control?

⇒ Concurrency control is the process of managing multiple transactions executing at the same time so that data remains consistent & conflicts are avoided. It ensures :-

- no data inconsistency.
- Proper execution order.
- safe multi-user access.

Example → Two users updating the same bank account should not overwrite each other's data.

Ques-4 What is Deadlock?

⇒ Deadlock is a situation where two or more transactions are waiting for each other's resources indefinitely & none of them can proceed.

Example :- ① Transaction T1 holds Resources A & waits for B.
② Transaction T2 holds Resources B & waits for A.

- Both are stuck forever.

Quick Work

Page No.
Date
M T W T F S S

Ques-5 What is serializability?

⇒ Serializability is a concept that ensures that the result of concurrent transactions is the same as if they were executed one by one. It helps maintain:-

- Data correctness.
- Consistency in concurrent systems.

Ques-6 What is Recovery in DBMS?

⇒ It is the process of restoring the database to a correct & consistent state after a failure (like system crash, power failure or error).

Techniques →

- Rollback (undo changes)
- Redo (reapply changes)

Ques-7 What is a checkpoint?

⇒ A checkpoint is a mechanism where the current state of the database is saved at a specific point in time.

Purpose →

- It reduces recovery time.
- It avoids re-executing all transaction.

Quick Work
Page No.:
Date:
M T W T F S S

Section-B

Ques-8 Explain ACID properties using Banking Example.

⇒ Scenario:- Money transfer from Account A → Account B

① Atomicity → Either both operations happen (debit → credit) or none.

↳ If system fails in between → transaction is cancelled.

Example → ₹1000 debited from A but not credited to B → rollback.

② Consistency → Database must remain correct before & after transaction.

Example :- Total money remain same.

Before = ₹10,000

After = ₹10,000

③ Isolation → Multiple transactions should not ~~Inf~~ interfere.

Example :- Two users transferring money at the same time should not see partial updates.

④ Durability → In this once transaction is committed, it is permanent.

Example :-

After transfer success, data remains saved even after system crash.

↳ ACID ensures secure, reliable & correct banking transactions.

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Quick Work
Page No.
Date:
M T W T F S S

Ques-9 Illustrate a deadlock situation with two transactions.

⇒ Scenario :- Transaction T1 locks Resource A.
- Transaction T2 locks Resource B.

Transaction

- T1 ⇒ Locks A → waiting for B
- T2 ⇒ Locks B → waiting for A.

- Both transaction wait forever so, it is a Deadlock situation.

Ques-10 Explain serial vs concurrent execution with example.

⇒

Serial	Concurrent
1) Transaction execution is one by one this is known as serial.	1) Transaction executes simultaneously this is known as concurrent.
2) In this transaction speed is slow.	2) In this transaction speed is fast.
3) Example :- T1 completes → then T2 starts.	3) Example :- T_1 & T_2 runs at the same time.

Section-C

Ques-11 A banking system transfers money b/w accounts. Explain how ACID properties apply.

Quick Work
Page No.:
Date:
M T W T F S S

Scenario → Transfer ₹ 1000 from Account A → Account B.

① Atomicity :-
- Both operations (debit from A + credit to B) must complete.
- If any step fails → entire transaction is cancelled (rollback).

② Consistency :-
- Database must remain valid before & after transaction.

Example :- Total balance remains same.
Before - ₹ 10,000
After - ₹ 10,000

③ Isolation :-
Transactions run independently.

Example :-
If two users transfer money at same time, they should not see each other's intermediate data.

④ Durability :-
Once transaction is committed, changes are permanent.

Example :
Even if system crashes, data remains saved.

Ques-12 Two transaction update same data simultaneously. Identify possible issues.

⇒ Concurrency Problems :- ① Lost update Problem
- In this one transaction overwrites another's update.

2.) Dirty Read :- A transaction reads uncommitted data.

3.) Non-Repeatable Read :- Same query gives different results.

Example :- T1 updates balance = 5000 → not committed

T2 reads this value → wrong data.

Soln → use locking mechanisms & apply isolation levels.

Ques-13 A system crashes during transaction. Explain recovery steps.

⇒ Transaction is incomplete → database may become inconsistent.

Recovery Steps → 1) Check Log File.
↳ DBMS maintains log of all operations.

2) UNDO (Rollback)
↳ Incomplete transactions are reversed.

3) REDO
↳ Completed transactions are reapplied.

4.) Restore from Checkpoint.
↳ Start recovery from last saved checkpoint

Example :- Transaction debited money but not credited.
↳ Rollback restores original balance.

Quick Work
Page No.:
Date:
M T W T F S S

Section-10

Ques-14 Explain one ACID property with real-life example.

⇒ **Atomicity:-** It means a transaction is treated as a single unit. - either all operations happen or none happen.

Real-life Example (Bank Transfer):-

- ₹ 1000 is transferred from Account A → Account B
- Step 1:- Debit ₹ 1000 from A.
- Step-2:- Credit ₹ 1000 ~~from~~ to B.

- If system fails after step 1 → money is lost. So, DBMS roll back the transaction.
- In this either both steps happen or none.

Ques-15 Draw a transaction timeline & identify conflict

⇒

Time	Transaction T1	Transaction T2
t_1	Read A	
t_2		Read A
t_3	Update A	
t_4		Update A

- Both T_1 & T_2 update the same data (A). This causes Write-Write conflict. This leads to lost update problem.

Ques-16 Explain one recovery technique.

⇒ Rollback is a recovery technique where incomplete or failed transaction are reversed to maintain consistency.